Gold Range Lab: The Math Behind the Models

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The math has two layers: the statistics measured on real gold futures (GC=F) candles, and the equations that drive the five physics-inspired animations.

Layer 1: Measured from the candles

Range and position. The range is `lo = min(Low)` and `hi = max(High)` over the last 288 bars. Position is `(price − lo) / (hi − lo)`, so 0 is the low and 1 is the high.

Volume profile and POC. Each bar's typical price is `(H + L + C) / 3`. These are binned into 48 slices of the range, weighted by volume, lightly smoothed, and normalised so they sum to 1. That gives `p(x)`, and the tallest bin is the point of control (POC).

Kramers index. `ln(p_POC / p_edge)`, where `p_edge` is the average of the 4 bins at that edge. A bigger number means the edge is much "thinner" than the centre, so it is a taller barrier in the analogy.

Volatility. Two high/low-based estimators:

- Parkinson: $\sigma = \sqrt{\text{mean}[\ln(H/L)^2] / (4 \ln 2)}$
- Garman-Klass: $\sigma = \sqrt{\text{mean}[\, 0.5 \cdot \ln(H/L)^2 - (2 \ln 2 - 1) \cdot \ln(C/O)^2 \,]}$
- Vol ratio: Parkinson over the last 24 bars divided by Parkinson over the whole range. Below 1 is calmer than the range average, above 1 is hotter.

CVD bias. $\Sigma\ \text{sign}(C - O) \cdot \text{Volume} / \Sigma\ \text{Volume}$ over the last 96 bars. It runs from −1 to +1. It is a candle-based proxy, not true bid/ask delta.

Permutation entropy.

1. Take 4 consecutive returns and record their ordering (24 possible patterns).
2. Count how often each pattern p_i appears over 96 returns.
3. Compute $S = -\Sigma\ p_i \ln p_i / \ln 24$.

0 means very orderly and 1 means pure noise. It measures randomness, not direction. The percentile is how many of the rolling values are at or below the current one.

Hurst exponent. Price changes over a lag `k` scale roughly as `std(P(t+k) − P(t)) ∝ k^H`. The code takes the slope of log(std) against log(lag) for lags up to 24:

- H below 0.5 means mean-reverting.
- H around 0.5 means random walk.
- H above 0.5 means persistent.

Derived scores.

- Compression: `clip(1 − vol_ratio, 0, 1)`
- Pressure: `0.55·compression + 0.45·min(1, 3·|CVD|)`
- Diffusion coefficient: `(std(returns) / ln(hi/lo))² / 2`

Regime rules. Expansion needs H above 0.58 and vol ratio above 1.2. Compression needs entropy percentile below 35% and vol ratio below 0.95. Anything else is "balanced".

Layer 2: The physics behind each animation

1. Potential well. Price follows the Langevin equation `dX = −∇U dt + √(2D) dW`. The potential is `U = −V₀ e^{−r²/2w²} − τx + …`, where `V₀` is the well depth, `w` its width and `τ` the order-flow tilt. The well floor is also ruffled by the volume profile, using `U = −D ln p`. The well disappears when `τ > V₀ / (w√e)`. Escape speed follows Kramers: `rate ≈ (ω/2π)·e^{−ΔU/D}`, so a lower barrier or more noise means exponentially faster escape.

2. Entropy. Each particle is pulled toward a lattice site by a spring `k`, with friction `γ` and thermal noise set by temperature `T`: `dv = [−k(x − x_site) − γv]dt + noise`. Higher `T` weakens `k`, so the lattice melts. The order parameter is `φ = mean(e^{−|x − x_site|²/a²})` and the disorder is `S = 1 − φ`.

3. Diffusion. Liquidity density `u` follows `∂u/∂t = ∇·(D·permeability·∇u) − v ∂u/∂x`. The range edges have near-zero permeability (about 0.02), so they reflect. Wall stress is the density jump across the wall. At the scripted "break" the permeability rises and a drift `v` carries the density outward.

4. Fluid. Particles feel a pressure force `−∇p` pushing from dense to sparse regions, plus drag `γ`. With the gate shut, drag keeps speed near zero while density rises. When it opens, a Bernoulli-style drive through the narrowing nozzle turns stored pressure into jet speed.

5. Fractal. Each filament's increments follow $dz_t = \varphi \cdot dz_{t-1} + \sigma\varepsilon$. For this process, $VR(2) = 1 + \varphi$, so $H = 0.5 \cdot (1 + \log_2(1 + \varphi))$, or inversely $\varphi = 2^{2H-1} - 1$. Negative φ gives tangled mean-reversion and positive φ gives trending filaments. The H shown on screen is measured on the simulated ensemble with the same lag-scaling estimator.

Limits to keep in mind

- The weights in the pressure score (0.55/0.45) and the regime thresholds are heuristics, not fitted or backtested.
- Hurst and entropy are noisy on short windows.
- The physics is analogy. Real price does not obey these equations, and the dashboard describes state without forecasting.